



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT PROPOSAL

SECJ2154

SECTION 02

OBJECT ORIENTED PROGRAMMING

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1. Introduction

Accurate and efficient staff attendance tracking is crucial in ensuring organizational productivity and discipline. Traditional manual methods, such as paper-based or spreadsheet systems, are prone to errors, data loss, and manipulation. In this project, we propose a Java-based **Staff Attendance Application** that allows full-time and part-time staff to clock in using the console. The system will log attendance with timestamps and store records in a file for administrative review. This project is designed using Object-Oriented Programming (OOP) principles to ensure modularity, code reuse, and maintainability.

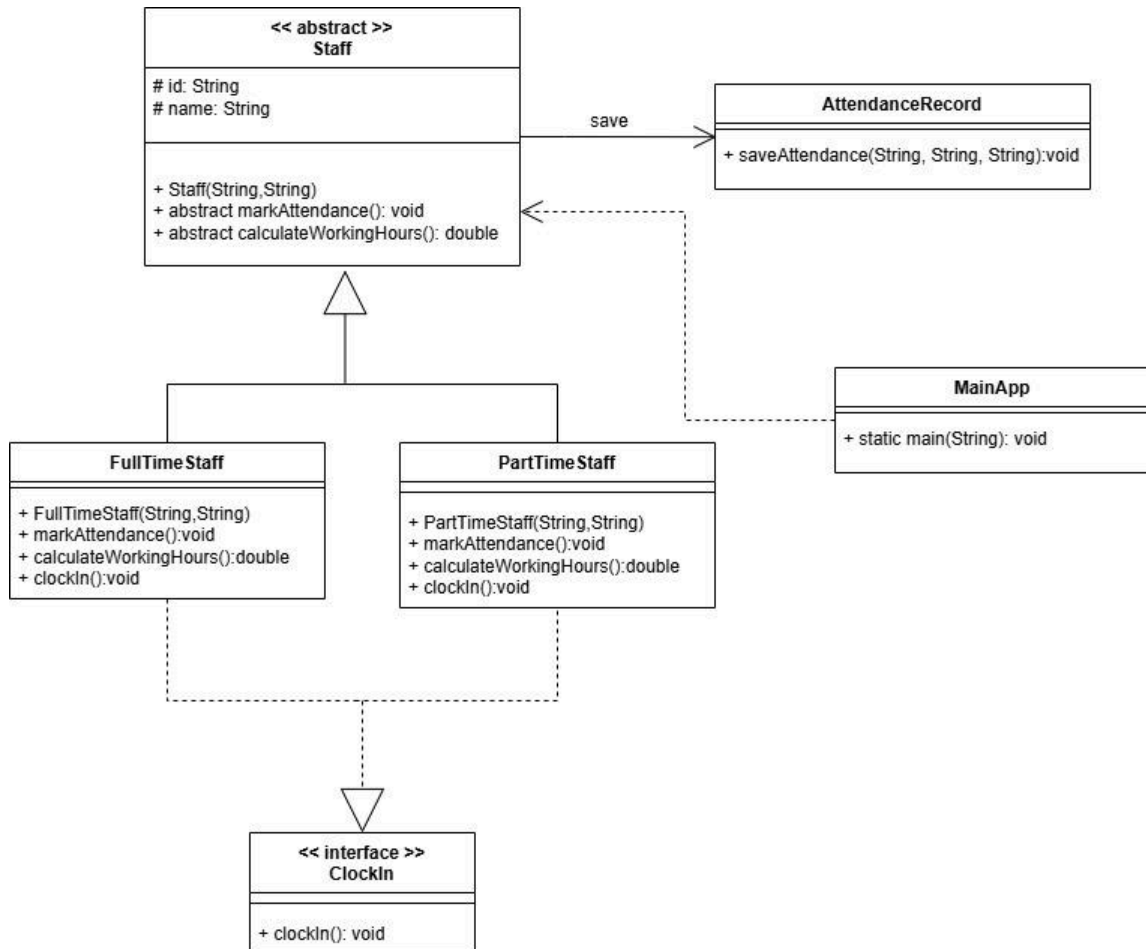
2. Problem Statement

Manual attendance tracking systems face several limitations:

- Risk of human error in logging attendance time
- Difficulty in maintaining attendance records over time
- No differentiation in handling full-time and part-time staff
- Time-consuming reporting and verification process

To overcome these issues, an automated Java application is proposed. The system enables staff to clock in via console input, and attendance records are saved with accurate timestamps in a file. This ensures a secure, consistent, and efficient way of tracking staff presence.

3. Class Diagram of the project



4. Techniques/Software/Tools Being Used

Main Concept of Java Programming to be used in this project:

- File
- Interface Class
- Exception Handling
- Abstract Class
- Inheritance
- Polymorphism
- Association

Tool used to draw class diagram: Drawio